Raj Nagar

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Professional Summary

Aspiring AI and Backend Engineer with a strong foundation in web architectures and decision-driven backend systems. Proficient in Python, FastAPI, Flask, and REST APIs, with hands-on experience across relational (SQL) and non-relational (MongoDB) databases. Experienced in implementing security-critical features like risk-based authentication, with growing expertise in LLMs, RAG, LangChain, and vector databases. Passionate about building secure, intelligent, AI-powered applications.

Education

MCA, SCSIT, DAVV, Indore	2025 - Present	CGPA: 7.57/10
BCA, Softvision College, Indore	2022 - 2025	CGPA: 8.4/10

Experience

Developer Intern at Changeforlife June 2026 - Present

- Built a browser-native Text-to-Speech feature using the Web Speech API with playback controls, progress tracking, and sentence highlighting.
- Developed a fully client-side TTS solution, eliminating server overhead while supporting multiple concurrent users.
- Enhanced application accessibility and user engagement by enabling in-browser audio playback without external TTS services.
- Integrated the feature into an existing PHP-based CMS while maintaining performance and cross-browser compatibility.

Technical Skills

Programming Languages: Python, Java, C++

Generative AI: LLMs, Chat Models, Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, AI Agents, Vector Databases (FAISS, ChromaDB)

Backend Development: FastAPI, Flask, REST APIs, SQLAlchemy, Pydantic, JSON

Databases: MySQL , PostgreSQL, MongoDB (Basic)

Machine Learning: Scikit-Learn, Pandas, NumPy, Data Preprocessing, Feature Engineering

Deep Learning: TensorFlow, Keras, CNN Fundamentals

Frontend: HTML, CSS, JavaScript (Basic), Bootstrap

Tools: Git, GitHub, Postman, VS Code, Jupyter Notebook

Projects

Trust Login Authentication System Python, FastAPI, JWT, MongoDB

Designed and built a risk-based authentication platform using FastAPI and MongoDB, going beyond standard login flows to make real-time trust decisions on each login attempt. Implemented JWT-based authentication with bcrypt password hashing (cost factor 12) and role-based access control, backed by a risk-scoring engine combining device fingerprinting, IP geolocation and proxy detection, and Google Safe Browsing checks. When calculated risk exceeds 50

Hotel Management System Flask, PostgreSQL (Supabase), SQLAlchemy, Bootstrap

Developed and deployed a full-stack hotel management system for my family's hotel business to streamline daily operations and replace manual record management. Built modules for room management, customer reservations, bookings, payments, and reviews using Flask and SQLAlchemy ORM with Supabase PostgreSQL. Designed a secure admin dashboard for managing hotel data, implemented user authentication and session management, and added Excel import/export functionality using Pandas and OpenPyXL for business reporting. Optimized database queries to improve application performance and responsiveness, reducing manual administrative work and enabling faster booking and customer management.

Customer Churn Prediction System Python, Scikit-Learn

Built a machine learning pipeline to predict customer churn using historical customer data. Performed data preprocessing, feature engineering, exploratory data analysis (EDA), and trained multiple classification models including Logistic Regression, Random Forest, and XGBoost. Improved model performance through hyperparameter tuning and feature selection, achieving an accuracy of **84%** . Evaluated the model using Accuracy, Precision, Recall, F1-Score, and ROC-AUC to ensure reliable customer churn prediction

Certifications

- HackerRank SQL (Intermediate)